In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FULL TEST – V

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

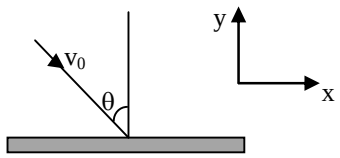
Physics

PART – I

SECTION – A
(Only One Option Correct Type)

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

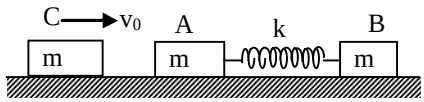
1. A ball is thrown such that it strikes the ground with velocity v_0 making an angle θ with the vertical. The height to which it rises after the collision (if the coefficient of restitution is e) is



(A) $\frac{e^2 v_0^2 \cos^2\theta}{2g}$ (B) $\frac{e^2 v_0^2 \sin^2\theta}{2g}$

(C) $\frac{e^2 v_0^2 \tan^2\theta}{2g}$ (D) $\frac{e^2 v_0^2 \cos^2\theta}{g}$

2. The maximum potential energy that the spring connecting blocks A and B can have as a consequence of collision of block C with A is (ignore any friction.)



(A) mv_0^2

(B) $\frac{mv_0^2}{2}$

(C) $\frac{mv_0^2}{4}$

(D) $\frac{2mv_0^2}{3}$

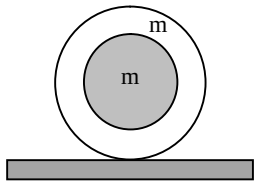
3. A block m_1 strikes a stationary blocks m_3 inelastically. Another block m_2 is kept on m_3. Neglecting the friction between all contacting surfaces, the maximum fractional decreases of K. E. of the system in collision is:

(A) $\frac{m_1}{m_1+m_2+m_3}$ (B) $\frac{m_1}{m_2+m_3}$

(C) $\frac{m_3}{m_1+m_3}$ (D) $\frac{m_2+m_3}{m_1+m_2+m_3}$

Space for rough work

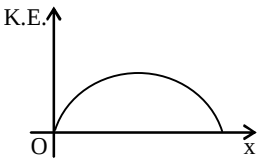
4. A thin hollow sphere of mass m is completely filled with an ideal liquid of mass m. when the sphere rolls with a velocity v kinetic energy of the system is equal to

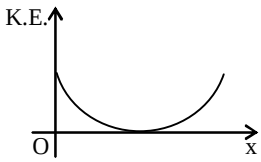


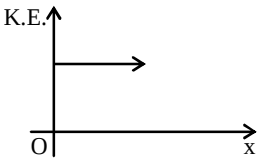
(A) $\frac{1}{2}mv^2$ (B) mv^2

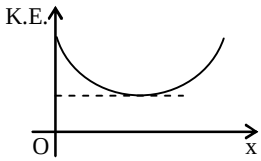
(C) $\frac{4}{3}mv^2$ (D) $\frac{4}{5}mv^2$

5. A projectile is fired with some velocity making certain angle with the horizontal. Which of the following graphs is the best representation for the kinetic energy of a projectile (K. E.) versus its horizontal displacement (x)?

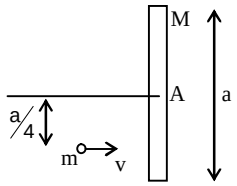
(A) 

(B) 

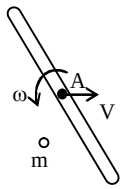
(C) 

(D) 

6. A uniform rod of mass M and length a lies on a smooth horizontal plane. A particle of mass m moving at a speed v perpendicular to the length of the rod strikes it at a distance of a/4 from the centre and stops after the collision. The angular velocity of the rod about its centre just after the collision is



Before Collision



After Collision

(A) $\frac{3mv}{Ma}$ (B) $\frac{2mv}{Ma}$

(C) $\frac{3Mv}{ma}$ (D) $\frac{4mv}{Ma}$

Space for rough work

7. Two coherent waves represented by $y_1 = A\sin\left(\frac{2\pi x_1}{\lambda} - \omega t + \frac{\pi}{6}\right)$ and $y_2 = A\sin\left(\frac{2\pi x_2}{\lambda} - \omega t + \frac{\pi}{8}\right)$ are superposed. Find the path difference $(x_1 - x_2)$ to produce constructive interference ($n \in I$).

(A) $n\lambda$ (B) $\left(n - \frac{1}{48}\right)\lambda$

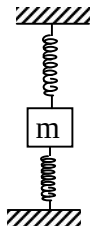
(C) $(n - 48)\lambda$ (D) $n\lambda / 2$

8. A small air bubble of radius r is found to form at a depth of H from the open surface of the liquid contained in a beaker. If S is the surface tension and ρ, the density of the liquid and p_0, the atmospheric pressure the pressure inside the bubble is

(A) $\frac{4S}{r} + \rho gH + p_0$ (B) $\frac{2S}{r} - \rho gH + p_0$

(C) $\frac{4S}{r} - \rho gH + p_0$ (D) $\frac{2S}{r} + \rho gH + p_0$

9. A block of mass m is suspended from a spring. Its frequency of oscillation is *f*. The spring is cut into two identical halves and the same block is suspended from one of the two pieces of the spring, such that it just touches the other spring below in its equilibrium position. The frequency of small oscillations of the mass will be



(A) f (B) $\sqrt{2}$ f

(C) 2f (D) $4f\left(\sqrt{2} - 1\right)$

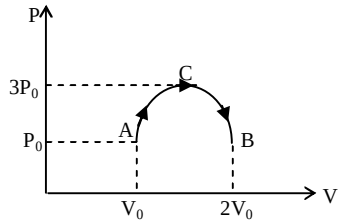
10. Two tuning forks with natural frequencies of 340 Hz each move relative to a stationary observer. One fork moves away from the observer white the other moves towards him at the same speed. The observer hears beats of frequency 3 Hz. The speed of the tuning fork is (velocity of sound = 340 m/s.)

(A) 1.5 m/s (B) 0.45 m/s

(C) 3 m/s (D) 4.5 m/s

Space for rough work

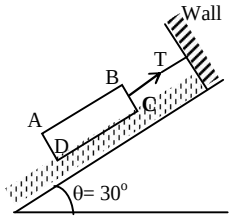
11. One mole of an ideal monatomic gas is taken through a semi-circular thermodynamic process ACB shown in the P–V diagram. The heat supplied to the system is



(A) $P_0^2\left(\frac{5+\pi}{2}\right)$ (B) $P_0V_0\left(\frac{2\pi-1}{2}\right)$

(C) $P_0V_0(1+\pi)$ (D) $P_0V_0\left(\frac{5+\pi}{2}\right)$

12. Water is flowing in a streamline down the incline having inclination 30° with the horizontal. A box ABCD, attached with the wall, is floating on the surface of the water keeping face CD of the box downward as shown in the figure. The box has a weight of 200 N and base are $1m^2$. If the velocity of the streamline at a distance of 1 mm from the base CD is 0.1 m/s, then the tension in the string connecting the block to the wall as shown is (Given viscosity of water = 0.1 N-s/m^2.]



(A) 110 N (B) 90 N

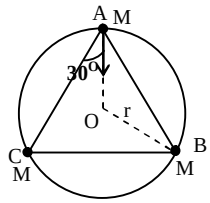
(C) 95 N (D) 100 N

13. The amplitude of a wave disturbance propagating in the positive x-direction is given by $y = 1/(1+x^2)$ at time t = 0 and by $y = 1/\left[1+(x-1)^2\right]$ at $t = 2$ seconds, where x and y are in metres. The shape of the wave disturbance does not change during the propagation. The velocity of the wave is

(A) $1\,ms^{-1}$ (B) $0.5\,ms^{-1}$

(C) $1.5\,ms^{-1}$ (D) $2\,ms^{-1}$

14. The particles each of mass M, are located at the vertices of an equilateral triangle of side 'a'. At what speed must they move if they all revolve under the influence of their gravitational force of attraction in a circular orbit circumscribing the triangle while still preserving the equilateral triangle?

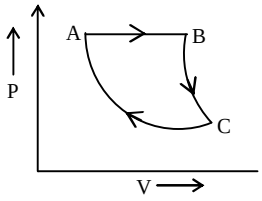


(A) $\sqrt{\frac{MG}{a}}$ (B) $\sqrt{\frac{2MG}{a}}$

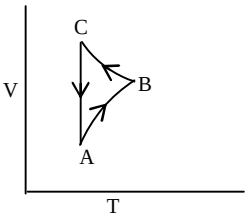
(C) $\sqrt{\frac{MG}{2a}}$ (D) None of the above

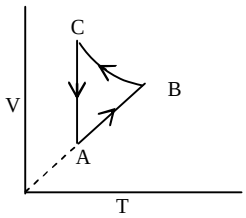
Space for rough work

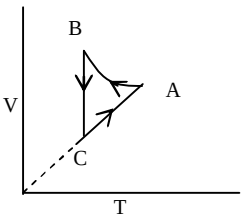
15. For a given thermodynamic process, the P –V diagram is as shown below: which of the following is the V – T diagram for the process?

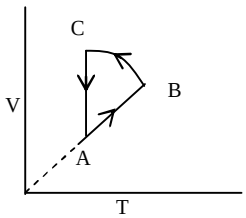


A → B : Isobaric
B → C : adiabatic
C → A : isothermal

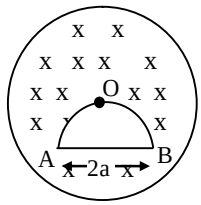
(A) 

(B) 

(C) 

(D) 

16. In cylindrical magnetic field B is changing as $B = B_0 + \alpha t$. Find the value of $\int \vec{E}.d\vec{l}$ for the path of BOA, as shown in the figure. Line AOB is semi circular. ($\vec{E}$ is the induced electric field in the region).



(A) $\frac{\pi a^2 \alpha}{2}$

(B) $a^2\alpha\left(\frac{\pi}{2}-1\right)$

(C) $a^2\alpha$

(D) $\pi a^2 \alpha$

Space for rough work

17. A capacitor of capacitance C is charged to a potential difference V from a cell and then disconnected from it. A charge +Q is now given to its positive plate. The potential difference across the capacitor is now

(A) V (B) $V+\frac{Q}{C}$

(C) $V+\frac{Q}{2C}$ (D) $V-\frac{Q}{C}$, if $V < CV$

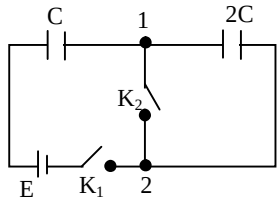
18. A potentiometer wire of length 100 cm has a resistance of 10Ω. It is connected in series with an external resistance and a cell of emf 2 volt and negligible internal resistance. A source of emf 10 mV is balanced against a length of 40 cm of the potentiometer wire. The value of external resistance is

(A) 800 Ω (B) 790 Ω
(C) 825 Ω (D) 745 Ω

19. An AC source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current resistered is I. If now the frequency of source is changed to ω/3 (but maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω will be.

(A) $\sqrt{\frac{3}{5}}$ (B) $\sqrt{\frac{5}{3}}$
(C) 3/5 (D) 5/3

20. Consider the capacitor circuit shown in the figure initially only K_1 is closed and after sufficiently long time, K_2 is also closed. The charge flowing through section 1 → 2 after closing the K_2 is

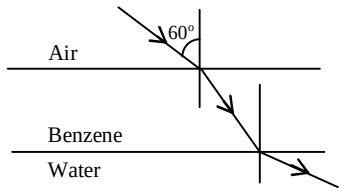


(A) $-\frac{CE}{2}$ from 2 to 1 (B) $-\frac{CE}{6}$ from 2 to 1

(C) $-CE$ from 2 to 1 (D) $\frac{CE}{4}$ from 1 to 1

Space for rough work

21. A layer of benzene floats on water. If the angle of incidence of the light entering benzene form air is 60°, then the angle that light makes with normal in water is
(Refractive index of water = 4/3.)



(A) 30°

(B) $\sin^{-1}\left(\frac{3\sqrt{3}}{8}\right)$

(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(D) data not sufficient

22. A ray of light is incident normally on one of the faces of a prism of apex angle 30° and refractive index $\sqrt{2}$. The angle of deviation of the ray is

(A) 15° (B) 75°

(C) 30° (D) none of these

23. Light of wavelength 4000 $\mathring{A}$ falls on a metal surface having work function of 2.0 eV. What is the stopping potential?

(A) 3.1 V (B) 2.0 V

(C) 5.1 V (D) 1.1 V

24. When the voltage applied to an X-ray tube is increased from V_1 = 10 kV to V_2 = 20 kV, the wavelength interval between the K_α line and the short-wave cut-off of the continuous X-ray spectrum increases by a factor η=3.0. Find the atomic number of the element of which the tube's anti-cathode is made.

(A) 29 (B) 30

(C) 28 (D) none of the above

25. Two radioactive materials X_1 and X_2 have decay constant 10λ and λ, respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be 1/e after a time

(A) $1/(10\lambda)$ (B) $1/(11\lambda)$

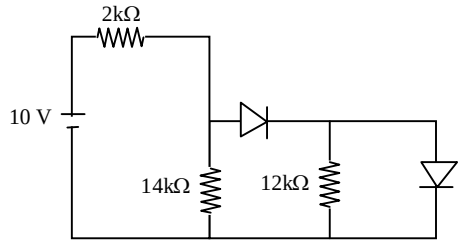
(C) $11/(10\lambda)$ (D) $1/(9\lambda)$

Space for rough work

26. In the measurement of the acceleration due to gravity using a simple pendulum, the percentage error in the measured value of the length of the pendulum is 2% and that in the time period of oscillation is 1%. What is the percentage error in the measured value of the acceleration due to gravity?
(A) 4% (B) 5%
(C) 6% (D) 3%

27. In a Coolidge tube, the potential difference used to accelerate the electrons is increased from 12.4 kV to 24.8 kV. As a result, the difference between the wavelengths of K_α line is $\left(\frac{hc}{e} = 12.4\,V\,\overset{o}{A}\right)$
(A) $1\,\overset{o}{A}$ (B) (B) $0.5\,\overset{o}{A}$
(C) $1.5\,\overset{o}{A}$ (D) 0

28. In the following circuit i_1 and i_2 are respectively (diodes are ideal)
(A) 0, 0
(B) 0, 5mA
(C) 5mA, 0
(D) 5mA, 5mA



29. For a common base configuration of PNP transistor $\frac{I_C}{I_E} = 0.98$ then maximum current gain in common emitter configuration will be
(A) 12 (B) 24
(C) 6 (D) 48

30. Due to annihilation of electron-positron of same kinetic energy 0.95 MeV, a photon is produced which can also be produced by an electron of energy E, the possible value (S) of E is e(mass of electron $C = 9.1\times10^{-31}$kg, $e = 1.6\times10^{-19}$c)
(A) 1.02 MeV (B) 2.42 MeV
(C) 1.03 MeV (D) 2.93 MeV

Space for rough work

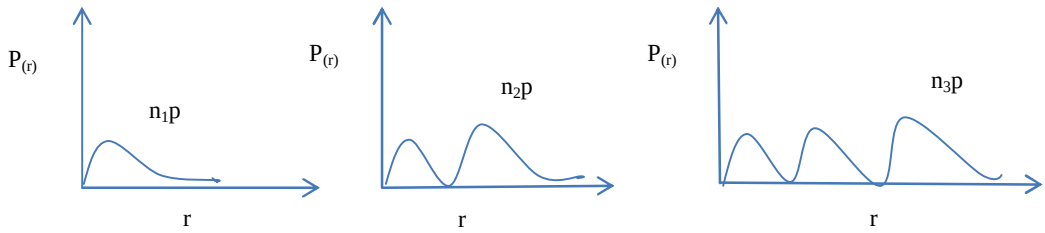
Chemistry

PART – II

SECTION – A
(Only One Option Correct Type)

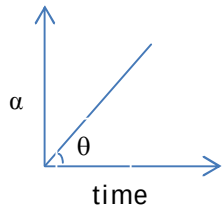
This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. When sum of radius ratio(using Bohr's modal) of hydrogen atom obtained from following given qualitative sketch of radial probability density $P_{(r)}$ vs r for p subshells is 'x'. Then value of $\frac{x-4\sigma}{2}$ (σ = value of bond order of c-c bond in benzene, and n_1, n_2, n_3 are corresponding shell number of p subshells whose sketches are given) is……



(A) 7
(B) 11.5
(C) 13.5
(D) 15.5

2. A zero order kinetic, 2A -------→ P. The initial concentration of reactant is 0.05M is given. The variation of degree of dissociation with time is …….

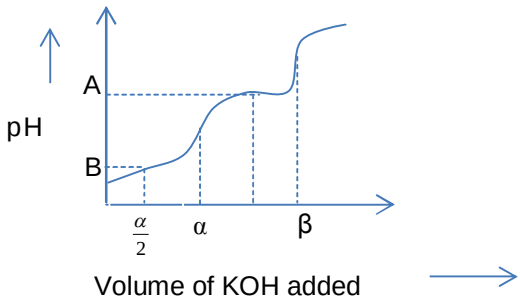


Using graph calculate value of rate constant . $\left(a = \text{degree of dissociation, } \tan\theta = 0.5\right)$.

(A) 0.125
(B) 0.0125
(C) 1.25
(D) 12.5

Space for rough work

3. The titration curve of diprotic acid with standard KOH solution is given as:



(A) The pH of amphiprotic anion represents at volume β.
(B) The pH (B) represent the 50% neutralization, pH that give idea about Pka_1
(C) pH of salt solution at 100% neutralization obtained at volume β.
(D) The pH (B) represents the end point at pH=7

4. $NaNO_3$ on heating to 500°C decomposes to produce $NaNO_2$ and oxygen, while salt on heating above 800°C, decomposes to Na_2O,N_2 and O_2 because of
(A) high heat of hydration of Na^+ ion than NO_3^-
(B) solvated electron of Na in $NaNO_3$ have high density than $NaNO_2$
(C) greater effective nuclear charge of Na^+ in $NaNO_3$ than in $NaNO_2$
(D) less polarizability of smaller nitrite ion than nitrate ion

5. The deposited mass of Metal M (molar mass = 50) on electrolysis of aq. MCl_x by passing 965 ampere current for 50 seconds is 5g.What will be the value of x+y+z of the following given ionic equation where M used as reducing agent.

$$xMnO_4^- + yM + zH^+ \longrightarrow aM^{+x} + bMn^{+2} + cH_2O$$

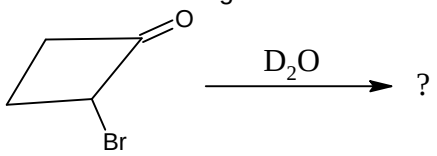
(A) 10 (B) 12
(C) 14 (D) 15

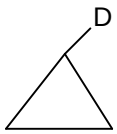
6. The P^H of 0.1 molar monobasic acid is 2.The osmotic pressure of solution at 25°C is
(A) 3 (B) 5
(C) 4 (D) 2.69

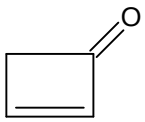
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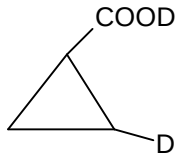
7. Which of the following will not give lassaigne test' for nitrogen?
(A) NH_2-NH_2 (B) C_6H_5-N=NC_6H_5
(C) CH_3CONH_2 (D)CH_3CN

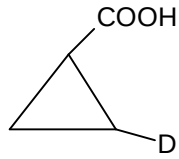
8. The Product of the given reaction:



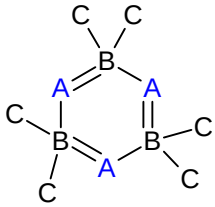
(A) 

(B) 

(C) 

(D) 

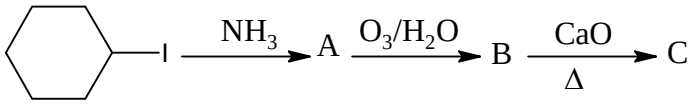
9. A stable molecule containing atoms of elements A,B and C has a following structure. Which of the following is possible combination of elements?

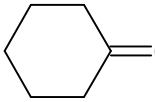
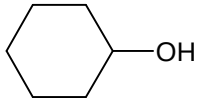
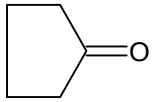
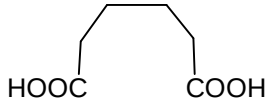


(A) N, P, Cl (B) O, S, Cl
(C) B, C, H (D) P, N, F

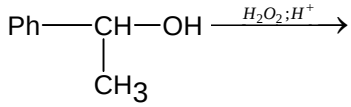
10. Which of the following statements are correct for S_N^2 reaction?
(A) In the gaseous phase, the order of reactivity of halide ions is $F^- > Cl^- > I^-$ while in methanol the order of reactivity is $I^- > Br > Cl^- > F^-$
(B) Among similar solvents, in general, the greater the polarity, slower the S_N^2 reaction.
(C) In an aprotic solvent such as dimethyl sulphoxide the order of reactivity of halide ions as nucleophiles is $F^{\Theta} > Cl^{\Theta} > Br^{\Theta} > I^{\Theta}$
(D) All are correct.

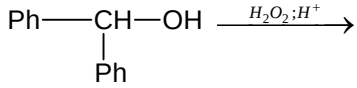
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11. 

(A)  (B)  (C)  (D) 

12. Which of the following reaction gives the alcohol, whose molecular weight is lowered by 14 unit than its starting reactant?

(A) $(CH_3)_2 CH - OH \xrightarrow{H_2O_2;H^+}$ (B) 

(C) $CH_3CH_2OH \xrightarrow{H_2O_2;H^{\oplus}}$ (D) 

13. Select the most effective drying agent at O°C out of the basis of following equilibria.

$$ACl_2 \cdot 6H_2O \rightleftharpoons ACl_2 \cdot 2H_2O + 4H_2O$$
(s) (s) (g)

$K_P = 6 \cdot 89 \times 10^{-12}$

$$A_2HPO_4 \cdot 12H_2O \rightleftharpoons Na_2HPO_4 \cdot 7H_2O + 5H_2O$$
(s) (s)

$K_P = 5.25 \times 10^{-13}$

$$ASO_4 \cdot 10H_2O \rightleftharpoons ASO_4 + 10H_2O$$
(s) (s) (g)

$K_P = 4.08 \times 10^{-25}$

The vapour Pressure of water at 0^0C is $4 \cdot 58$ torr in atmosphere?

(A) $ACl_2 \cdot 2H_2O$ (B) $AHPO_4 \cdot 7H_2O$ (s)

(C) ASO_4 (D) All are equal

Space for rough work

14. $Cr_2O_7^{-2} \xrightarrow{P^H = X} CrO_4^{-2} \xrightarrow{P^H = Y} Cr_2O_7^{-2}$

Select the correct option.

(A) X > Y (B) X < Y

(C) X = Y (D) X = Y = 7

15. Statements:

I. An especially favorable conformation for the peptide linkages in protein is the '$\propto$' helix arrangement.

II. I_4O_9 ionises as $I^{3+} + \left(IO_3^-\right)_3$

III. O_2 reacts with P_tF_6 to give the compound $\left[O_2^+\right]\ \left[P_tF_6\right]^-$,

IV. Froth floatation process used for concentration of sulphide ore where sodium ethyl xanthate used as collector. Out of these statements,

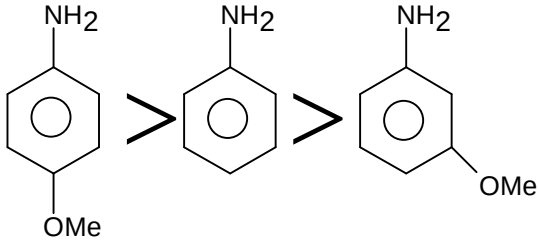
(A) I, II and III are correct. (B) II, III, and IV are correct

(C) Only III correct (D) I, II, III and IV are correct.

16. Which of the following order(s) of properties given is/are correct?

I. Tendency of alkyl halides to undergoes dehydrohalogenation in williamson, synthesis Follows the order $3^0 > 2^0 > 1^0$

II. Propanol < 2 – methyl – 1 – Propanol < 2 – butanol < 2 – Methyl – 2 Propanol (rate of reaction with HBr)

III. 

(Basic order)

IV. Toluene < m-dichlorobenzene < O- dichloro benzene < P- dichloro benzene - (Order of dipole moment).

Select correct option:

(A) I, II, IV, are correct (B) IV,I are correct

(C) II, II, IV, are correct (D) I, II, and III, are correct

Space for rough work

17. The specific conductivity of $AB_{(aq)}$ at 298 k is 1.26×10^{-6} and ionic conductivity of $A^{\oplus}_{(aq)}$ and $B^{\ominus}_{(aq)}$ at infinite dilution are 62 s cm^2 mol^{-1} and 76 s cm^2 mol^{-1} respectively. The number of $B^{\ominus}$ ions in saturated solution per litre (aproximately)
(A) 1×10^{14} (B) 5×10^{-14}
(C) 55×10^{14} (D) 1×10^{24}

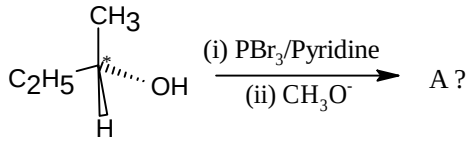
18. S_8 exist as a mixture of S_8 and S_2 according to dissociation $S_8 \rightleftharpoons 4S_2$
The rate of diffusion of mixture is 0.75 times more than that of pure sulpher(S_8). The mass percentage of S_2 in the sample
(A) 40 % (B) 30 %
(C) 80 % (D) 52 %

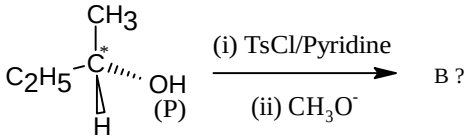
19. In silicon dioxide
(A) There are double bonds between silicon and oxygen atoms
(B) silicon atom is bonded to two oxygen atoms
(C) Each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded two silicon atoms.
(D) Each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms.

20. Statement not correct regarding ortho and para hydrogen
(A) Ortho is more stable than Para
(B) Ortho and Para hydrogen are called spin isomers.
(C) magnetic moment of Para is zero while ortho has twice that of proton.
(D) Conductivity of ortho and Para is equal.

21. Which of the following mixtures can be separated by using an NH_3 solution?
(A) Fe^{+3} and Al^{+3} (B) Al^{+3} and Zn^{+2}
(C) Sn^{+2} and Pb^{+3} (D) Cu^{+2} Cd^{+2}

Space for rough work

22. 



Select the correct statements:
(A) A and B have same configuration
(B) A and B are diastereomers with each other
(C) A have same configuration as the starting material
(D) A and B have same configuration as the starting material

23. Incorrect statement is:
(A) $HCHO \xrightarrow[K]{\bar{O}H} HCOO^- + CH_3OH$, The reaction can have rate law expression,
$r = K\left[HCHO\right]^2\left[\bar{O}H\right]$ or $r = K\left[HCHO\right]^2\left[\bar{O}H\right]^2$
(B) ortho chlorobenzaldehyde and meta chlorobenzaldehycle gives cannizarro reaction.
(C)  $+H_2N-OH \xrightarrow{P^H\ 3.4to4} A \xrightarrow{SOCl_2} B,$
where B is an amide.
(D) acid halide > acid anhydride > aldehydo > ketone > ester $\simeq$ carboxylic acid > amide > carboxylate ion. (relative reaclivities towards Nuclcophile)

24. The group of negative charged colloids
(A) starch, Haemoglobin and ferric hydroxide
(B) Au, Ag and Aluminium hydroxide
(C) Clay Haemoglobin and starch
(D) Arsenic sulphide, starch, and clay

Space for rough work

25. The incorrect statement among the following is

(A) $\left[Ni\left(CN\right)_4\right]^{2-}$ is square Planer complex

(B) $\left[ZnCl_4\right]^{2-}$ is tetrahedral complex

(C) In $\left[Cu\left(NO_2\right)_4\right]^{4-}$ tetragonal distortion takes place.

(D) The bond length of C – O' in CO (free state) is more than in complex Fe $(CO)_5$ (as a ligand)

26. Of the following mixture, the most likely, mixture nearer to ideal solution is

(A) NaCl – water
(B) ethanol – benzene
(C) heptane – water
(D) heptane - octane

27. A clourless salt changes to yellow on heating, salt is also soluble in NaOH as well in dil HCl. Salt can be

(A) FeO
(B) PbO
(C) ZnO
(D) CdO

28. The temperature voltage coefficient for a simple alkaline torch battery is - 6×10^{-4} VK^{-1}
What is the entropy change associated with battery discharge? The number of electrons transferred in the cell reaction n = 2.

(A) -1.16 JK^{-1} mol^{-1}
(B) 1.16 JK^{-1}
(C) 11. 6 JK^{-1} mol^{-1}
(D) -11.6 JK^{-1} mol^{-1}

29. Vander wall constant for hydrogen chloride gas a = 3.67 atm $litre^{-2}$ and b = .0408 mL mol^{-1}. The ratio of Tc (critical temperature) and Pc (critical pressure) of the gas is (approximately)

(A) 2
(B) 3
(C) 4
(D) 5

30. How many moles of HIO_4 (Periodic acid) are required for cleavage C- C bonds in the sucrose?

(A) 5
(B) 2
(C) 8
(D) 3

Space for rough work

Mathematics

PART – III

SECTION – A
(Only One Option Correct Type)

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. The value of integral $\int_0^2 \cos^{-1}\left\{(x-1)|x-1|+(x-1)^3\right\}dx$ is

(A) 2π (B) 3π
(C) 4π (D) π

2. If $\ln\left(2x^2+a^2\right) > \ln\left(x^2+2x+3\right)$, then set of values of 'a' will be

(A) $(-\infty,-2)\cup(2,\infty)$ (B) $(-\infty,-1)\cup(0,\infty)$
(C) $(-2,\infty)$ (D) R

3. f(x) = $\sin\left(\pi\left\{\frac{x}{2}\right\}\right)$, $x \in R$.

Then
(A) f(x) is discontinuous where x = 2K, $k \in I$.
(B) f(x) is discontinuous where x = 2K, $k \in I^+$
(C) f(x) is discontinuous every where except x = 2
(D) f(x) is continuous every where.

4. If $\sqrt{x^2+x+1} \geq 2-x$ exhaustive set of x is

(A) $(3/5,\ \infty)$ (B) $[3/5,\ \infty)$
(C) $[3/5,\ 2)$ (D) $[2,\ \infty)$

5. If $f(x)=\frac{x+1}{2x-1}$, then $f(f(x))$ is

(A) x (B) x + 1
(C) x – 1 (D) None of these

Space for rough work

6. If $f(x) = \max\left(3 - x^3, |x-3|\right)$ then f(x) is non differentiable at exactly
(A) 4 point (B) 2 point
(C) 3 point (D) 1 point

7. If in a triangle ABC, $b\cos^2\frac{A}{2} + a\cos^2\frac{B}{2} = \frac{3c}{2}$, then a, b, c are in
(A) GP (B) AP
(C) HP (D) None of these

8. If $\tan B = \frac{5\sin A\cos A}{1-5\cos^2 A}$, then the value of $4\tan(A+B) + \tan A$ is
(A) -1 (B) 2
(C) 0 (D) 4

9. If $\sin 3\theta = 4\sin\theta\sin 2\theta\sin 4\theta,\ 0 \le \theta \le \pi$, then the number of real solution for 'θ' is
(A) 2 (B) 6
(C) 0 (D) 8

10. In a triangle ABC, I is incentre and I_1, I_2, I_3 are excentres, then the value of $(II_1)\cdot(II_2)\cdot(II_3)$ is
(A) Rr^2 (B) $2R^2r$
(C) $2Rr^2$ (D) $16R^2r$

11. If $\cos\alpha = \frac{2\cos\beta - 1}{2\cos\beta}$, when $\theta < \alpha, \beta < \pi$ and $\tan\theta = \left|\frac{\tan\alpha/2}{\tan\beta/2}\right|$, then θ is
(A) 60^0 (B) 45^0
(C) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (D) 30^0

12. If $2x - 2y + 3z = 0,\ x + ay + 2z = 0$ and $2x + az = 0$ have a non-trivial solution, then
(A) $a = 1$ (B) $a = 2$
(C) $a = 0$ (D) None of these

Space for rough work

13. $\frac{a\cos A+b\cos B+c\cos C}{a+b+c}$ is

(A) $\frac{r}{R}$ (B) $\frac{R}{2r}$

(C) $\frac{R}{r}$ (D) $\frac{2r}{R}$

14. If $a\cos^3\alpha+3a\cos\alpha\cdot\sin^2\alpha=m$ and $a\sin^3\alpha+3a\cos^2\alpha\sin\alpha=n$ then value of $(m+n)^{2/3}+(m-n)^{2/3}$ is

(A) a^2 (B) $2a^{2/3}$

(C) $2a^{1/3}$ (D) $2a^3$

15. The coefficient of x^r where $o\le r\le n-1$ in the expression $(x+2)^{n-1}+(x+2)^{n-2}(x+1)+(x+2)^{n-3}(x+1)^2+.....+(x+1)^{n-1}$ is

(A) ${}^nC_r\left(2^r-1\right)$ (B) ${}^nC_r\left(2^{n-r}-1\right)$

(C) ${}^nC_r\left(2^r+1\right)$ (D) ${}^nC_r\left(2^{n-r}+1\right)$

16. If the term independent of 'x' in the expansion $\left(x\sin\theta+\frac{\cos\theta}{x}\right)^{10}$ is $f(\theta)$, then value of $f\left(\frac{\pi}{12}\right)$ is

(A) ${}^{10}C_5$ (B) $2^{-10}\,{}^{10}C_5$

(C) $2^{-5}\,{}^{10}C_5$ (D) $2^5\,{}^{10}C_5$

17. The sum of last three digit if the number $N = 7^{100} - 3^{100}$ is

(A) 0 (B) 1

(C) 3 (D) 4

Space for rough work

18. If A + B = $\frac{\pi}{2}$, then

(A) $(1-\tan A/2)(1-\tan B/2)=2$ (B) $(1+\cot A/2)(1+\cot B/2)=2$

(C) $(1+\sec A/2)(1+\cos ec\, A/2)=2$ (D) $(1+\cot A/2)(1+\cot B/2)=2$

19. If $a_1, a_2, a_3 ..... a_{20}$ are in AP and $a_1+a_4+a_7=10$ and $a_2+a_{10}=8$ the sum of all terms is

(A) 110 (B) 120

(C) $\frac{110}{3}$ (D) 220

20. If the points $A(x_1,y_1), B(x_2,y_2), C(x_3,y_3)$ lie on the parabola $y^2=2x^2$ such that x_1, x_2, x_3 are in A.P. and y_1, y_2, y_3 are in G.P. If the common ratio has two values r_1 and r_2, then $r_1 + r_2$ is

(A) 6 (B) $4\sqrt{2}$

(C) 8 (D) – 6

21. If $a_1, a_2, a_3 .... a_n$ is an increasing GP and $a_1^2+a_2^2+a_3^2=91$ also $a_1+a_2+a_3=13$, then S_n is

(A) $\frac{3^n-3}{2}$ (B) $\frac{3^{n+1}-3}{2}$

(C) 3^n-3 (D) $\frac{3^{n+1}-3}{4}$

22. If the sum of the roots of the quadratic equation $ax^2+bx+c=0$ is equal to sum of the square of their reciprocal, then, a/c, b/a, c/b are in

(A) AP (B) GP

(C) HP (D) none of these

23. A variable rectangle PQRS has its sides parallel to fixed directions, Q and S lie respectively on the lines x = a, x = - a and P lies on the x-axis. Then locus of R is

(A) circle (B) parabola

(C) straight locus (D) Hyperbola

Space for rough work

24. From an external point P, pair of tangents are drawn to the parabola $y^2 = 4x$, α and β are the inclination tangents with x axis such that $\alpha + \beta = \frac{\pi}{4}$. The locus of P is

(A) $x - y - 2 = 0$ (B) $x - y - 1 = 0$

(C) $2x + 3y - 1 = 0$ (D) $x - 3y + 1 = 0$

25. For an ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with vertices A and A', tangents are drawn at the point P in the first quadrant meets the y-axis in Q and chord A'P meets y-axis in M. If O is the origin, then $OQ^2 - MQ^2$ is equals to

(A) 4 (B) 7

(C) 5 (D) 2

26. Let $y^2 = 4ax$ and circle $x^2 + y^2 + 2bx = 0$ touch each other externally then the equations $ax^2 + cx - b = 0$ will have

(A) Both roots are real and distinct all for all $c \in R$

(B) Both roots are complex for all $\in R$

(C) Both roots are real and distinct $\forall c > ab$

(D) Both roots are complex $\forall c < ab$

27. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola xy = c^2 in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then

(A) $x_1 + x_2 + x_3 + x_4 = c^2$ (B) $x_1 + x_2 + x_3 + x_4 = -c^2$

(C) $x_1 + x_2 + x_3 + x_4 = 0$ (D) $y_1 + y_2 + y_3 + y_4 = c^2$

Space for rough work

28. If A is a orthogonal matrix of order 4 and det (A) = 2, then value of det (A - I) is.
(A) 1 (B) -1
(C) 0 (D) 2

29. Genereal solution of $\frac{dy}{dy}=\frac{x^2+y^2+1}{2xy}$ is
(A) $y^2+1=x^2+cy$ (B) $y^2+1=x^2+cx$
(C) $x^2+1=y^2+cy$ (D) $x^2+1=y^2+cx$

30. A fair coin is tossed 'n' number of times. The probability that head will turn up an even number of times, is equal to
(A) $\frac{n-1}{2n}$ (B) $\frac{1}{2}$
(C) $\frac{n+1}{2n}$ (D) $\frac{2^{n-1}-1}{2^n}$

Space for rough work